

SKILL WEEK 9

Task 1:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {    int fd =
open("input.txt", O_RDONLY);    if (fd <
0) {        perror("input.txt");
return 1;    }

    dup2(fd, STDIN_FILENO);
close(fd);

    char buf[100];    ssize_t n = read(STDIN_FILENO,
buf, sizeof(buf) - 1);    if (n < 0) {
perror("read");        return 1;    }

    buf[n] = '\0';    printf("Input from
redirected file:\n%s", buf);    return 0; }
```

OUTPUT:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill9.1.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill9.1.c -o skill9.1
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill9.1
Input from redirected file:
rcb
```

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Task 2:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {    int fd = open("output.txt", O_WRONLY | O_CREAT |
O_TRUNC, 0644);    if (fd < 0) {        perror("output.txt");
return 1;    }

    dup2(fd, STDOUT_FILENO);
close(fd);

    printf("This output is redirected to a file.\n");
printf("Output redirection completed.\n");

    return 0;
}
```

OUTPUT:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill9.2.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill9.2.c -o skill9.2
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill9.2
This output is redirected to a file.
Output redirection completed.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```

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Task 3:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {    int fd = open("append.txt", O_WRONLY | O_CREAT |
O_APPEND, 0644);    if (fd < 0) {        perror("append.txt");
return 1;    }

    dup2(fd, STDOUT_FILENO);
close(fd);

    printf("New line appended to the file.\n");
return 0; }
```

OUTPUT:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill9.3.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill9.3.c -o skill9.3
^[[Askjuneeth@LAPTOP-D8SJ2A02:~$ ./skill9.3
Enter command using >> : echo Second line >> output.txt
Append operation completed successfully.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```

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Task 4:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {    int fd = open("error.txt", O_WRONLY | O_CREAT |
O_TRUNC, 0644);    if (fd < 0)        return 1;

    dup2(fd, STDERR_FILENO);
close(fd);

    fprintf(stderr, "Error: file could not be opened.\n");
fprintf(stderr, "stderr redirection completed.\n");

    return 0;
}
```

OUTPUT:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill9.4.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill9.4.c -o skill9.4
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill9.4
Program started...
error.txt opened successfully.
Error messages were redirected to error.txt
Program completed successfully.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```